

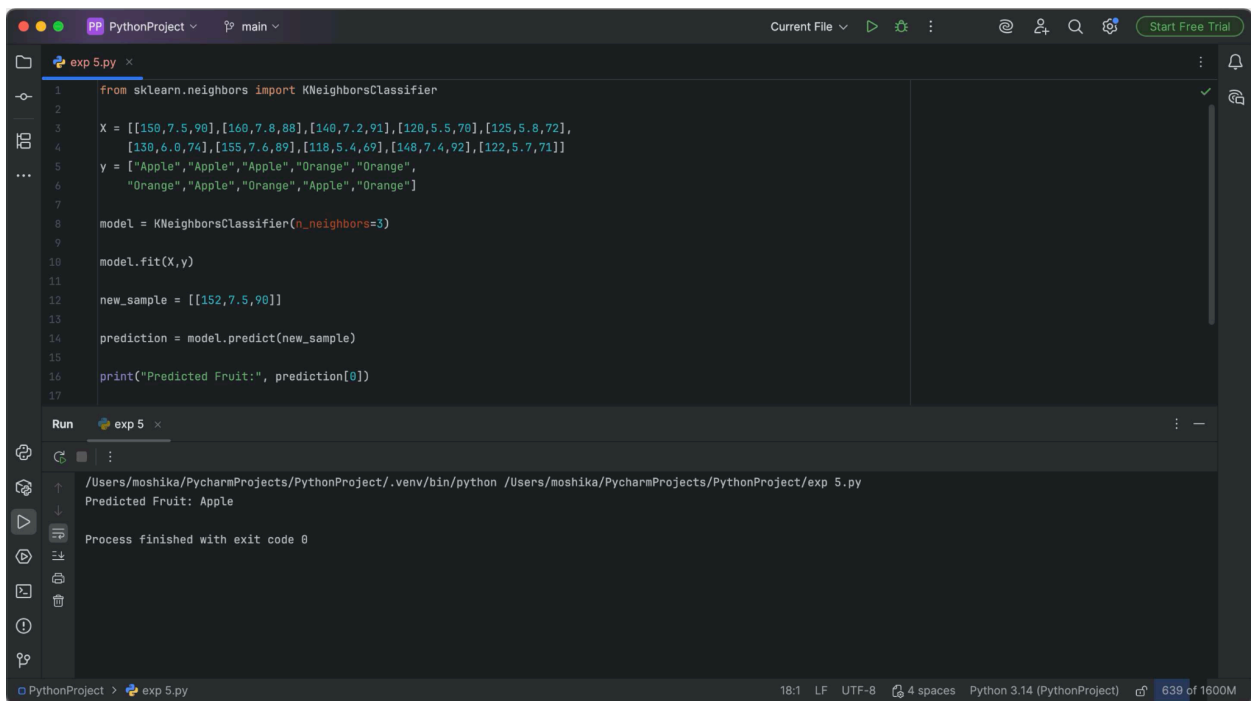
EXPERIMENT - 5

Aim: To implement the K-Nearest Neighbours algorithm in Python for classifying a given dataset.

Procedure:

1. Load the dataset.
2. Separate input features and target labels.
3. Split the data into training and testing sets.
4. Choose the value of K.
5. Calculate the distance between the test sample and training samples.
6. Select the K nearest samples.
7. Count the class labels of the nearest samples.
8. Assign the majority class to the test sample.
9. Repeat for all test samples.
10. Calculate and display the classification accuracy.

Output:



```
1 from sklearn.neighbors import KNeighborsClassifier
2
3 X = [[150, 7.5, 90], [160, 7.8, 88], [140, 7.2, 91], [120, 5.5, 70], [125, 5.8, 72],
4      [130, 6.0, 74], [155, 7.6, 89], [118, 5.4, 69], [148, 7.4, 92], [122, 5.7, 71]]
5
6 y = ["Apple", "Apple", "Apple", "Orange", "Orange",
7      "Orange", "Apple", "Orange", "Apple", "Orange"]
8
9 model = KNeighborsClassifier(n_neighbors=3)
10
11 model.fit(X, y)
12
13 new_sample = [[152, 7.5, 90]]
14
15 prediction = model.predict(new_sample)
16
17 print("Predicted Fruit:", prediction[0])
```

Run exp 5

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 5.py
Predicted Fruit: Apple
Process finished with exit code 0
```

Result: The K-NN algorithm successfully classifies the test samples based on their nearest neighbours and gives the predicted class labels.